

From explicit sparse codebooks to generated supports

Preserving the broad $L = 1$ model class while attacking storage and inverse search

Stage	5 of 5
Updated	9 September 2026
Status	Jobs 027 and 029 complete; job 028 partial; amplitude job 030 pending on HPC

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in **results/**.

1 Research philosophy at the current frontier

The strongest empirical lead remains the global $L = 1$ codebook, not the sectioned outer-code route. The desired method should therefore remove the exponential implementation cost without unnecessarily replacing that broad family by one specific tree or parity construction. This is a methodological principle rather than a claim that restrictions can always be avoided:

1. begin from the unrestricted explicit object whose behaviour is understood;
2. identify redundancy or invariance empirically and algebraically;
3. impose the smallest tractable structure that removes the bottleneck;
4. compare that restriction against the broader parent family before adding another;
5. distinguish compact forward generation from scalable inverse decoding.

If a restriction is eventually necessary, it should enter because an experiment or proof justifies it, not because a named construction makes implementation convenient.

2 The explicit result to preserve

A B -bit payload has $M = 2^B$ possible values. In the explicit one-component model,

$$\Phi = [\phi_0, \dots, \phi_{M-1}] \in \mathbb{R}^{n \times M}, \quad y = \Phi a + z = \sum_{m=0}^{M-1} a_m \phi_m + z, \quad \|\phi_m\|_2 = 1.$$

Job 027 fixed $B = 12, n = 256$ and swept exact support size

$$s \in \{256, 192, 128, 96, 64, 48, 32, 24, 16, 12, 8, 6, 4, 3, 2, 1\}.$$

Within each seed, supports were nested prefixes of a random ordering and amplitudes were paired before renormalisation. Dense and four-mask ODMA controls used the same load/SNR contract.

The strict 72-summary aggregate gives representative high-SNR PUPE:

Family/support	Density $s/256$	D0	D1
Dense	1	.2935	.2030
Sparse global $s = 64$	.25	.2932	.2256
Sparse global $s = 48$	.1875	.2975	.2188
Sparse global $s = 32$	.125	.3051	.2276
Sparse global $s = 24$	.09375	.3120	.2399
Sparse global $s = 16$	.0625	.3312	.2323
Sparse global $s = 8$	.03125	.3984	.2813
Sparse global $s = 4$	.015625	.5171	.3994
Four-mask ODMA	.25	.5363	.4884

Table 1: Job 027 mean PUPE over 8/12 dB and $K_a = 9, 17, 26, 30$. Lower is better.

D0 remains nearly flat through $s \approx 48$; at $s = 32$ it is only 0.0116 above dense. D1 at $s = 16$ is 0.0293 above dense in the strict aggregate, but D1 had only 20 epochs versus 80 for D0 and shows more two-seed noise. At equal density $s = 64$, arbitrary sparse global beats ODMA by 0.2431 PUPE for D0 and 0.2628 for D1.

The repaired seed-2702 full-density endpoints are not a clean nested paired comparison because resampling one exact-zero Gaussian draw changed random-number consumption. The intermediate supports retain the intended pairing and carry the main frontier conclusion.

Evidence status: Measured job 027. It demonstrates a forgiving explicit sparse-global model class under current D0/D1. It does not remove 2^B storage or scoring, and D1 is not established as optimally trained.

3 Why counting available supports is insufficient

There are $\binom{n}{s}$ unsigned size- s masks. For $n = 256$,

$$\log_2 \binom{256}{21} \gtrsim 100, \quad \log_2 \binom{256}{24} \approx 111.36.$$

Thus support labels alone can in principle distinguish 2^{100} messages around $s = 21$ – 24 . This does not solve the communication problem. A useful family must also have balanced rows, controlled pair intersections and signed correlations, separation between sums of K_a columns, and a receiver that can propose likely messages without enumerating all 2^B labels.

Job 027 shows this distinction directly. All 4096 support/sign patterns remain distinct through $s = 4$, but performance degrades much earlier. As s falls from 32 to 8, sampled correlation q99.9 rises from about 0.256 to 0.479 and the fraction of rows occupied by a $K_a = 30$ active set falls from 0.982 to 0.613. At $s = 1$, the namespace itself finally collapses: only 256 supports and 512 real signed directions exist.

The conservative procedural target $s = 24$ is therefore a rate/geometry starting point, not a $B = 100$ feasibility claim. $s = 21$ is an aggressive combinatorial boundary. Neither number supplies an inverse algorithm.

4 Generator–proposer specification

Write a message as $w \in \{0, 1\}^B$. A procedural compact encoder replaces stored columns by

$$S_\theta(w) = \{\rho_0(w), \dots, \rho_{s-1}(w)\},$$

$$\phi_\theta(w) = \sum_{t=0}^{s-1} \alpha_t(w) e_{\rho_t(w)}, \quad \sum_t |\alpha_t(w)|^2 = 1.$$

The payload bits are binary inputs to the generator; the transmitted nonzero values α_t may be real or complex.

Forward generation is only half of a scalable code:

$$\text{codeword}_\theta(w) \longrightarrow \phi_\theta(w),$$

$$\text{pro}(y) \longrightarrow \mathcal{C}(y), \quad |\mathcal{C}(y)| \ll 2^B,$$

where the candidate list should contain the active messages with high recall. Distinct generated descriptions provide isolated-message injectivity only. The inverse search must work in noise and in a multiuser superposition; otherwise the receiver still tries 2^B messages.

The desired tests are therefore not only PUPE. They include proposal recall, candidate-list size, runtime/memory, unit-energy error, row balance, support repetition, pair overlap/correlation, active-set Gram spectra, K -sum separation, and complete-message recovery after the proposer.

5 Master table/bin hash skeleton

Let $n = TR$: divide resources into T disjoint tables with R rows each. Choose one row in every table:

$$\rho_t(w) = tR + h_t(w), \quad h_t(w) \in \{0, \dots, R-1\}.$$

With phase/sign decoration $\xi_t(w)$,

$$g(w) = \frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} \xi_t(w) e_{tR+h_t(w)}.$$

If $|\xi_t(w)| = 1$, then $\|g(w)\|_2^2 = 1$ exactly. Two codewords have inner product

$$\langle g(w), g(v) \rangle = \frac{1}{T} \sum_{t=0}^{T-1} \mathbf{1}\{h_t(w) = h_t(v)\} \xi_t(w)^* \xi_t(v).$$

Hashes control overlap and balance; phases can cancel the signed sum. The restriction changes the support family from all $\binom{n}{T}$ masks to R^T one-per-table masks. Its rate condition is

$$B \leq T \log_2 R = T \log_2(n/T).$$

Expected pair overlap under uniform table choices is $T/R = T^2/n$, matching the first-order value for two independent size- T supports. Tail geometry can still differ, so it must be measured against arbitrary sparse supports.

6 Binary affine hash family

When $R = 2^r$, let

$$h_t(w) = A_t w \oplus b_t, \quad A_t \in \mathbb{F}_2^{r \times B}, \quad b_t \in \mathbb{F}_2^r.$$

$\mathbb{F}_2$ means arithmetic on bits: addition is XOR. Full row rank of each A_t makes every table exactly balanced. Full column rank of the stacked matrix

$$A_{\text{stack}} = [A_0^T \cdots A_{T-1}^T]^T \in \mathbb{F}_2^{Tr \times B}$$

makes the complete hash tuple injective, because

$$A_{\text{stack}} w = A_{\text{stack}} v \implies A_{\text{stack}}(w \oplus v) = 0 \implies w = v.$$

For difference $d = w \oplus v$, the two supports collide in table t exactly when $A_t d = 0$. At small B , enumerating all nonzero d gives the exact overlap spectrum. Job 028 selects the best of 128 random full-rank banks by worst and tail collision counts.

The matrices A_t, b_t are fixed discrete design parameters, not gradient-trained. They can be randomly sampled, searched, or designed algebraically. Keeping their rank constraints exact is more important than forcing them into a continuous relaxation. The phase/amplitude rule and downstream decoder can still be learned.

7 Connection back to the explicit factorisation

At $B = 14$, every generated support can be materialised without changing D0/D1. Define

$$(P_t)_{jm} = \mathbf{1}\{j = h_t(w_m)\}, \quad D_t = \text{diag}(\alpha_{t0}, \dots, \alpha_{t,M-1}).$$

Then

$$\Phi = \begin{bmatrix} P_0 D_0 \\ P_1 D_1 \\ \vdots \\ P_{T-1} D_{T-1} \end{bmatrix} \in \mathbb{R}^{TR \times 2^B}, \quad \Phi_{tR+j,m} = \alpha_{tm} \mathbf{1}\{j = h_t(w_m)\}.$$

This is an $L = 1$ global matrix with generated support, not the section-count representation of Report 4. D0/D1 retain their unchanged residual and adjoint operations. The small- B experiment therefore asks a clean question: does this support restriction damage the promising explicit landscape before a new proposer is built?

8 What is learnable without breaking the skeleton

If $S \in \{0, 1\}^{n \times M}$ is the fixed materialised support and V contains trainable amplitudes,

$$\Phi = S \odot V, \quad \frac{\partial \mathcal{L}}{\partial V} = S \odot \frac{\partial \mathcal{L}}{\partial \Phi}.$$

The mask makes all off-support gradients zero. Projection after every optimiser step,

$$\phi_m \leftarrow \frac{\phi_m}{\|\phi_m\|_2},$$

guarantees unit power for the final learned codebook. Dense controls use the same projection.

The original certification stores amplitudes per message. To remove that second exponential object, choose one invertible $Q \in \mathbb{F}_2^{B \times B}$ and let P_J be its first J rows. Define

$$u_J(w) = P_J w \in \mathbb{F}_2^J, \quad V \in \mathbb{R}^{T \times 2^J}, \quad \alpha_J(w) = \frac{V_{:,u_J(w)}}{\|V_{:,u_J(w)}\|_2}.$$

The transmitted column is

$$\phi_J(w) = \sum_{t=0}^{T-1} \alpha_{J,t}(w) e_{tR+h_t(w)}, \quad \|\phi_J(w)\|_2^2 = 1.$$

Thus the saved generator contains A, b, P_J, V and needs $T2^J$ amplitude values rather than $T2^B$. At $J = 0$, all messages share one learned table profile. At $J = B$, $P_B = Q$ is bijective, so V assigns an arbitrary amplitude vector to every message and recovers the old model class. Because the P_J are prefixes, any J -bit bank can be copied into the two halves of a $(J + 1)$ -bit bank; hence these classes are genuinely nested.

The support maps A, b and label map P_J remain fixed discrete designs. In a learned run, gradients from every message with label u accumulate into $V_{:,u}$ through the column normalisation; in a fixed run, V is a buffer and only the decoder trains. This gives a paired test of amplitude adaptation without changing support. It also guarantees deployment power for messages never sampled in training. A selected-message $B = 100, n = 256, T = 64, J = 8$ regression test generates columns without allocating a 2^{100} object. Current D0/D1 still allocate such a message axis, so forward scalability does not establish inverse scalability.

9 Jobs 028–030 and their causal decision gates

Job 028 fixes $B = 14, n = 256, T \in \{16, 32\}$, D0/D1, and two seeds. It compares:

1. iid arbitrary sparse support;
2. balanced random one-per-table support;
3. random affine binary hashes;
4. collision-spectrum-selected affine hashes;
5. dense context.

The sequence isolates the cost of tables, then linear/XOR structure, then offline hash selection. Fixed Gaussian amplitudes are paired across sparse families and normalised exactly.

Job 029 jointly trains amplitudes and the chosen decoder for 120 epochs on dense, iid sparse, and selected-hash supports at $T = 16, 32$. It records pre/post geometry and full loss curves. This separates a fixed-amplitude or under-training floor from a limitation of the support skeleton. It is still a small- B model-class test because the learned amplitudes and D0/D1 states are message-indexed. All 20 rows returned. Selected hash and iid sparse differ by less than 0.011 PUPE in every matched aggregate, so no important hash penalty is visible at current precision. Fixed and joint runs used different training budgets, so they do not isolate the value of amplitude learning; job 030 makes that comparison fair.

Job 030 fixes the selected $T = 32$ hash and sweeps $J \in \{0, 2, 4, 8, 14\}$ at $B = 14, n = 256$. Every J has a fixed- V row and a matched jointly learned- V row for D0/D1 and two seeds. Fixed equal-amplitude and Rademacher controls are included. All rows have a 120-epoch ceiling, stop after five validation epochs without improvement, and restore the best checkpoint. This isolates amplitude model-class loss before candidate proposal is introduced.

The decision order is deliberately conservative:

1. if balanced tables are materially worse than iid sparse, reject or revise the table restriction;
2. if tables survive but affine hashes fail, retain the skeleton and change the generator family;
3. if selected hashes match iid support, find the smallest J that matches the $J = B$ amplitude endpoint;
4. only then introduce a proposer and measure proposal recall separately from final PUPE;
5. keep exact small- B materialisation as a permanent regression reference.

Evidence status: Local tests verify exact support size, unit energy, $\mathbb{F}_2$ ranks, support injectivity, nested amplitude classes, the unrestricted $J = B$ endpoint, gradients only into V , and finite fixed/joint D0/D1 paths. A 12-run $B = 8$ mini-grid improved held-out loss in every tested case. No job-030 HPC performance result is recorded yet.

10 Current claim and boundary

The project has evidence that a broad sparse-global $L = 1$ family can be much sparser than dense without reproducing the ODMA penalty. It has a compact candidate support skeleton whose forward map is injective and unit-energy by construction. It now has a compact nested amplitude family, but no HPC evidence yet showing which J preserves the unrestricted endpoint, and no high-recall sub-exhaustive inverse proposer for noisy multiuser sums.

That boundary is the research opportunity. The next successful design should look like the favourable explicit global model to the channel, but like a compact generator and searchable algebra to the implementation. This preserves the general direction without mistaking storage compression, isolated-message invertibility, or local section decoding for complete scalable URA.